

## Topic 7 Review [98 marks]

1. [Maximum mark: 15]

25M.1.HL.TZ2.11

Each set of traffic lights in a city is controlled by a separate embedded system.

(a.i) Describe what is meant by an embedded system.

[2]

### Markscheme

*Award **max** [2].*

An embedded system is a microprocessor/controller-based system within a larger system or device;  
which is designed to perform a specific task/can be programmable/have a fixed functionality;

### Examiners report

Most responses were aptly written with the correct description of embedded system. The absence of correct keywords made some responses vague and lacking in technical clarity.

(a.ii) State **two** other examples of embedded systems in daily life.

[2]

### Markscheme

*Award **max** [2].*

Industrial machines;  
Household appliances;  
Automobiles/airplanes;  
Medical equipment;

Digital watches;  
Cell phones;  
Toys;

**Note:** *Reward other reasonable answers.*

#### Examiners report

A variety of embedded systems were mentioned in the responses. However, repetition of examples related to household appliances limited some students to only half the available marks. Additionally, many students referred to 'street lights' without specifying 'automated streetlights'—the inclusion of the term 'automated' would have added the necessary clarity and accuracy to their answers.

It has been proposed that all the traffic lights in the city should be controlled from a central computer.

- (b) Discuss the advantages **and** disadvantages of using one central computer to manage the city's traffic light system.

[4]

#### Markscheme

*Award **max** [4].*

*Award max [2] for pros and max [2] for cons.*

*Pros:*

Better control over traffic flow/any problem appearing at one traffic light is known immediately/can be dealt with immediately;

Traffic lights can be quickly adapted to avoid traffic blocks/when levels of traffic flow change;

All traffic lights can be synchronized efficiently, allowing smooth traffic flow across the city (e.g. creating green waves on main roads);

Easier to add additional hardware/inputs such as sensors;  
Cheaper-reduces the need for separate hardware (controllers or advanced systems) at each intersection/cheaper maintenance;  
Centralised data collection (accidents, traffic) that could help with future planning;  
Control program can be done at one place, with different programs for different times of day and related traffic patterns;  
easier maintenance/software updates can be performed centrally without having to make adjustments at every intersection;

*Cons:*

single point of failure – failure of the central computer stops the entire city's traffic system;  
system depends on high-speed communication links, network failure can disrupt activity;  
the capacity/speed of the central computer can be limited/cannot provide the desired levels of service (many inputs with differing priorities which could take time);  
high initial cost of communication system/central control system/software;  
security risks/cyberattacks on the central computer affects the entire city;  
reduced local autonomy/individual intersections cannot make local decisions like dealing with an unexpected blockage/emergency;  
difficult to scale up as the city expands (system may struggle to handle increased load)/disruption in service/increased complexity in decision-making and processing;

**Note:** *Reward other reasonable benefits and drawbacks.*

### Examiners report

Students often overlooked that the question required at least two advantages and disadvantages. Several responses lacked clarity, with vague points such as 'easier to control'—without specifying what is easier to control or outlining how. In fact, such a claim can be misleading, as

managing the logic for coordinating all traffic lights in a city can be quite complex.

Vehicle counts provide the city's traffic control centre with information regarding the number of vehicles on the city's roads.

- (c) List **two** types of sensor that could be used to collect data on the number of vehicles.

[2]

#### Markscheme

*Award **max** [2].*

Motion sensors;  
Pneumatic tubes;  
Piezoelectric sensor;  
Inductive loop;  
Magnetic sensors;  
Acoustic detectors;  
Infrared detectors/sensors;  
Doppler sensors;  
Radar/microwave/ultrasonic sensors;  
Pressure sensors;  
Image sensors;  
Radio frequency sensors;  
Proximity sensors;

**Note:** *Reward other correct answers.*

#### Examiners report

Students identified correct sensors in their responses. It is recommended that, in future exams, they select the most intuitive and commonly used

sensors to demonstrate clearer understanding and relevance.

(d) Describe **one** difference between interrupts and polling.

[2]

#### Markscheme

*Award **max** [2].*

Interrupt can occur at any point in time (an interrupt is asynchronous);  
processor polls the devices at regular intervals of time (polling is a synchronous activity);

Interrupt occurs only when a device requires servicing;  
Polling occurs when processor needs to wait and check if a device needs to be serviced/processor continuously checks the status of the device to find whether it requires attention;

Interrupts save the processor cycles;  
polling wastes many of the processor cycles;

Interrupts are priority based;  
Polling involves sequential checking with no priority;

Interrupts allow the CPU to respond quickly to a situation;  
Response may be delayed depending on polling order/frequency;

Polling involves creating a software loop (to regularly and continuously read sensor data);  
Interrupt is a hardware mechanism (in which the sensor notices processor that it requires its attention);

#### Examiners report

Responses that included definitions of interrupt and polling generally scored full marks. Some students effectively highlighted the key differences between the two techniques.

The data about traffic and road conditions will be shared with application (app) developers.

Their apps can provide real-time information about traffic conditions.

- (e) Explain the benefit for drivers of having access to real-time information about traffic and road conditions.

[3]

#### Markscheme

*Award **max** [3].*

*Award [1] mark for benefit,*

*Award [1] for situation/example*

*Award [1] for elaboration/justification*

Example answers;

to reduce accidents/improving safety;  
by optimizing flows across intersections;  
during periods when roads are shut down due to construction/any other events;

enables the optimization of traffic flow;  
saving time/fuel/cost;  
which is of great benefit for taxi drivers, logistics providers, on-demand services such as food delivery;

early warnings about traffic incidents;  
improve driver safety/save lives;  
a quick passage to emergency vehicles (ambulances/fire tenders) can be facilitated;

providing traffic advisories/alternative route diversion;  
improve drivers' comfort;  
which means less stress/less traffic accidents;

insights into congestion/delays across the city road network;  
enable traffic management authorities to create smarter mobility solutions;  
that reduce pollutant emissions that can cause respiratory problems/other health issues;

insights into congestion/delays/traffic accidents;  
allows drivers to delay or prepone the trip (based on current traffic trends);  
plan the departure time more effectively;

**Note:** *Reward other reasonable answers.*

#### Examiners report

Nearly all students explained the benefits for drivers comprehensively covering points for safety, optimization, comfort, etc.

2. [Maximum mark: 2]

24N.1.HL.TZ0.8

Outline **one** advantage of using robots in the manufacturing of cars.

[2]

#### Markscheme

*Award [2 max]*

*Award [1] for identifying the advantage and award [1] for stating why robots provide this advantage.*

Improved quality of cars;  
robots are more precise/ reliable than humans;

Increased safety;

Using robots for boring/ repetitive tasks means fewer risks of injury for workers/ robots can work in hazardous conditions (painting, assembling parts);

Reduced costs;

No salaries required/ healthcare savings/ insurance savings;

Increased speed for manufacturing processes;

robots work at constant speed/ don't need breaks / robots operate 24/ ;

**Note:** *Reward other reasonable answers.*

#### Examiners report

Many students outlined correctly one advantage of using robots in the manufacturing of cars.

3. [Maximum mark: 2]

24M.1.HL.TZ1.10

A temperature sensor is used in an automatic washing machine to help maintain the temperature of the water.

Outline the use of **one other** type of sensor used in an automatic washing machine.

[2]

#### Markscheme

*Award [2 max]*

*Award [1] for type of sensor and award [1] for appropriate extension.*

Water level sensor/ ultrasonic sensor;

To switch water supply on/off as required / to make sure sufficient water is in the machine/keep door locked until water drains;

Proximity sensor;

To make sure the door is closed before starting the machine;

Vibration sensor;

To stop the drum from spinning if the load is unbalanced;

Speed (of rotation) sensor;

To make sure the drum is rotating at the correct speed of the wash cycle;

Load/ weight sensor;

to measure the weight of laundry and adjust the washing parameters;

Pressure sensor;

to monitor the water pressure and water levels inside the tub;

Moisture sensor;

to control auto-dry cycles/ how dry the clothes are;

Dirt sensor;

Detects soil and detergent levels and adjusts the washing cycle;

**Note:** Sensor type must be identified to get a mark for extension.

### Examiners report

Quite a few students repeated the 'temperature sensor' in their answers. There was also a lot of guess work on this. Others who were able to score full two marks here identified the name of sensor correctly along with an appropriate description.

4. [Maximum mark: 4]

24M.1.HL.TZ1.11

Describe the role of feedback in a system that uses sensors and a microprocessor to control the temperature of a room.

[4]

#### Markscheme

*Award [4 max]*

Input to the system is provided by sensors/ temperature sensor;  
Sensors continuously send digitised data to microprocessor/ ADC converts data collected from sensors to digital data and sends it to the microprocessor;  
The microprocessor would then compare the input readings against stored values / required temperature settings;  
The microprocessor would then calculate any error value/adjusting factor required for the heating / cooling equipment;  
This difference /error value is fed back to the microprocessor to perform a suitable action (by heater/cooler)/ Error value is fed back as an input to the system;  
The above steps are repeated until the system is turned off which maintains the temperature of a room;

#### Examiners report

There were very small number of completely correct answers to this question with many students achieving at least one or two marks for the mention of 'sensors detecting temperature' or 'the use of microprocessor to compare detected temperature with set values'. Several students incorrectly mentioned "sensors detect changes in temperature" and lost marks.

5. [Maximum mark: 15]

23N.1.HL.TZ0.12

Input devices that detect cars approaching a crossroads are connected to a microprocessor.

- (a.i) Identify **two** types of sensor that can be used to detect approaching cars.

[2]

#### Markscheme

*Award [2 max].*

Electromagnetic sensors;

Pressure sensors;

Infrared sensors;

Ultrasonic sensors;

Microwave sensors;

Inductive loop detectors;

Acoustic sensors;

#### Examiners report

Terminology is not always used in an appropriate and competent way and these questions are often approached by providing superficial common knowledge

The vast majority of students correctly identified two types of sensors that can be used to detect approaching cars.

- (a.ii) Outline why sensors are appropriate input devices in this situation.

[2]

#### Markscheme

*Award [2 max].*

Sensors are appropriate because they automatically sense the presence in the given environment;

without the need for human operation/sensors can detect 24x7 without losing efficiency;

Sensors are appropriate because only presence of a car required; this can be reliably detected by sensors/data is sent to CPU quickly enabling quicker reaction times;

Sensors are appropriate because it saves staffing the crossroads; so, costs are reduced;

- (b) Suggest the type of memory that could be used to store the control program in the microprocessor.

[2]

#### Markscheme

*Award [2 max].*

ROM;

Because it is non-volatile;

PROM;

can be programmed only once and is not erasable;

EPROM;

Can be reprogrammed in case of change in traffic flow;

EEPROM;

can be erased one byte at a time (rather than erasing the entire chip)/allows flexible process of reprogramming;

#### Examiners report

A mixed set of responses was seen for this question.

The traffic lights at the crossroads are also connected to a microprocessor. A person who wishes to cross the road presses a button at a traffic light. This causes an interrupt.

(c.i) Outline what is meant by an interrupt.

[2]

Markscheme

*Award [2 max].*

An interrupt is a signal sent to the processor;  
that halts the current process;

Examiners report

Many students provided answers that were vague or too general.

(c.ii) Explain how the microprocessor can deal with this interrupt.

[3]

Markscheme

*Award [3 max].*

The microprocessor responds to that interrupt with an interrupt handler/an ISR (Interrupt Service Routine)/a routine which instructs the microprocessor on how to handle the interrupt;  
the routine can send a signal to output transducers;  
to immediately turn the traffic lights to red to stop the cars;

The microprocessor receives an interrupt request and passes control to the interrupt handler;  
it may wait a certain amount of time before sending the signal to output transducer;  
to turn the traffic lights to red to stop the cars;

Cameras are installed on the top of the traffic lights at the crossroads.

(d.i) Outline **one** benefit of monitoring the traffic with cameras.

[2]

### Markscheme

*Award [2 max].*

Cameras improve traffic flow;  
by controlling the movement of vehicles and determining traffic light timing;

Traffic cameras help to reduce accidents;  
by controlling the movement of vehicles/identifying drivers who speed;

Cameras improve safety/have a preventive/disciplining effect;  
can be used to identify defaulters and to take legal action/make them pay fine;

Installation and deployment of automated traffic cameras often pays for itself in the long run;  
is still far less expensive than physical policing/traffic cameras can operate tirelessly 24/7;

Traffic cameras can be deployed with very little cost to infrastructure;  
as they can run independently from the electrical grid only on solar power;

### Examiners report

Most students provided generally correct answers. Students lost marks by not providing answers with sufficient clarity.

(d.ii) Outline **one** concern about monitoring the traffic with cameras.

[2]

### Markscheme

*Award [2 max].*

Privacy is an issue;  
as traffic cameras enable monitoring the activities of people (drivers, pedestrians);

It can be costly;  
to install/maintain a large number of cameras and monitoring systems;

Cameras can be vulnerable to damage;  
For example, wetness, moisture;

Cameras can be vulnerable to misuse;  
for example, criminals/hackers might have worked out ways to disable or  
disconnect cameras;

6. [Maximum mark: 15]

23M.1.HL.TZ1.12

Smart control systems can manage the temperature within a house.

- (a) Outline the steps involving the sensor, processor and output transducer to manage the temperature in the house.

[5]

#### Markscheme

***Award [5 max]***

a pre-set value of temperature stored/ user inputs a desired temperature;  
sensor measures/detects temperature;  
and it (regularly) sends the temperature readings to the microprocessor;  
microprocessor compares the actual readings with the pre-set/inputted temperature;  
if the temperature is above or below the pre-set temperature/ too hot or too cold;  
then the microprocessor sends signal to output transducer to adjust temperature accordingly;  
These steps are (continuously) repeated in a cycle;

***Note:*** Award [1 max] for AD or/and DA conversion, if explained in the response.

#### Examiners report

The majority of students attempted it and many got it right. However, some responses were vague enough not to describe the situation with reference to the basic technical terminology of the devices.

(b) Describe the role of feedback in this control system.

[2]

#### Markscheme

##### ***Award [2 max]***

A feedback system samples the output signals (temperature) then a sample (a fraction of the output signal) is fed back to the input;  
to change/modify the input signal;  
in such a way as to produce an output signal that differs substantially from the output signal produced in the absence of such feedback / as to increase or decrease the temperature in the house;

The output value is (continuously) compared to the desired value to produce an error value/ difference/ adjusting factor;  
the error value/ difference/ adjusting factor is fed to input;  
to determine the next action/ to achieve a desired outcome/pre-set temperature in the room;

#### Examiners report

The majority of students attempted it and many got it right. However, some responses were vague enough not to describe the situation with reference to the basic technical terminology of the devices.

(c) The smart control system is managed by an operating system.

(c.i) Describe **one** function of an operating system.

[2]

#### Markscheme

*Award [2 max]*

*Award [1] for identifying a function of an OS, and [1] for a description.*

*Award marks for **any** of the OS functions, for example, memory management, processor management, device management, coordination between other software and users, security, control over system performance, file management, etc.*

For example

Processor Management/Scheduling;

the OS decides the order in which processes have access to the processor /  
how much processing time each process has/ keeps tracks of the status of  
processes/ allocates the CPU to a process/ de-allocates processor when a  
process is no more required;

Examiners report

Many candidates provided vague or too general answers.

(c.ii) Outline **one** reason why a dedicated operating system would  
be used.

[2]

Markscheme

*Award [2 max]*

**Note: Accept reasons such as security, customisability, size, speed.**

Because a dedicated operating system is custom made/ designed  
specifically for the computer system/ for the available hardware;  
to do a specific function at maximum efficiency/ makes the most efficient  
use of the available resources;

Because the OS is specifically made to manage the control system/ for cell  
phone (that does not require to do any general-purpose tasks); meaning  
that it is faster than universal OS;

## Examiners report

Many candidates provided vague or too general answers.

- (d) Compare and contrast a centralized control system with a distributed control system for managing the temperature of a house.

[4]

## Markscheme

### *Award [4 max]*

*Award marks for giving an account of similarities and differences between centralized system and distributed system.*

### *Award [2 max] for centralized system.*

Easier to use- it allows control at a central point;

Controlled by a single processor;

Centralized system is easy to physically secure;

More cost-efficient for small systems (such as a house);

Large systems are usually more expensive to install due to the connectivity required between the various components;

Highly dependent on the network connectivity (the system can fail if the nodes lose connectivity as there is only one central node);

If it must be expanded (new equipment and processes added) downtime is usually required;

When the software program must be modified it requires downtime/ can have inadvertent effects on the existing control scheme;

When the central server is down, no other entity is there to send/receive responses/requests;

In case of a local failure (usually) the whole system stops working;

### *Award [2 max] for distributed system.*

Not beneficial to build and operate small distributed systems because of the low cost/benefit ratio;

Controlled by a collection of independent processors;

Distributed system is usually easier to install due to fewer connection issues;  
Easier to modify/upgrade without disturbing other processes;  
Allows individual control of settings at each unit or room, so it is more flexible;  
More complicated to control remotely;  
Difficult to know which node failed/ which node responded;  
In case of a local failure, all other processes continue to work;

Examiners report

Many candidates provided vague or too general answers.

7. [Maximum mark: 15]

22N.1.HL.TZ0.12

A company is planning to transform its office building into a smart building.  
Among other things, a smart building can control the opening and closing of its doors.

- (a) Outline **two** other operations that can be controlled by a smart building.

[4]

Markscheme

*Award [4] max.*

Heating / air conditioning controlled;  
help lower the energy bills / increases the comfort level of the occupants;

Monitoring hot water system;  
temperature and pump control monitoring via the building monitoring system allows for a proper functioning of hot water distribution through the facility;

Security system;

Locking/unlocking doors for the authorised or unauthorised employees/  
windows open-close;

Sprinkler system;

Monitored for fire safety;

Monitoring environment;

to know whether we can take advantage of it;

*Accept examples, such as, lights can be turned off if outdoor luminosity is high enough for indoor working;*

Electrical system;

the consumed electrical power and the state of main electrical switches can  
be controlled/monitored;

Monitoring/controlling pressure/humidity;

and if their value exceeds defined limits, an alarm is generated;

*Mark as [2] and [2]*

**Note:** Reward other suitable answers.

#### Examiners report

Generally, well answered, with many full or near full marks awarded.

Sensors, processors and output transducers are vital components of a smart building. They play an important role in the collection and management of data.

- (b) Explain how a smart building can control the opening and closing of all its doors. You should refer to sensors, processors and output transducers.

[4]

#### Markscheme

*Award [4] max.*

electromagnetic sensors detect if a door is opened/closed;

*(Accept other appropriate type of sensors! For example, proximity sensor detects when a person comes close to the door, movement sensor detects people approaching the door.)*

the read/detected sensor data is transmitted to processor;

processor determines whether any action should be taken (if a change in the state is detected-action needed);

then processor sends the signal (about which door to close/open/how long to keep it open) to the output transducer/ actuator;

which opens/closes the door:

the whole process is continually repeated;

#### Examiners report

Well answered. Some answers, however, contained incorrect/imprecise sentences (for example, confusion between the role of a processor and a sensor).

An operating system has a significant role in a smart building system.

(c.i) Identify **two** functions of this operating system.

[2]

#### Markscheme

*Award [2] max.*

Managing peripherals;

Managing memory;

Resource allocation;

**Note:** Reward other correct responses.

#### Examiners report

Surprisingly, some candidates had difficulty identifying two functions of the operating system, but in general the question was answered well.

- (c.ii) Suggest **one** technique this operating system might use to determine when a hardware device needs attention.

[2]

Markscheme

*Award [2] max.*

Interrupt;

An input device sends a signal which causes the processor and the OS to stop the current operation and determine what to do next;

OR

Polling;

All input devices are continuously checked by the processor to see whether they want to communicate;

- (d) Explain why transforming the building into a smart building will be beneficial for the company.

[3]

Markscheme

*Award [3] max.*

*Award marks for the benefits such as energy savings, productivity gains, reducing heating costs, increase the productivity of the facility staff, improve building operations, support sustainability efforts; enhance decision-making across the organization, etc.*

energy savings;

the use of optimal start/stop;

which allows the building automation system to learn, for example, when it should bring the air conditioning system online (for a particular zone in the building);

Improved asset performance;

which reduces energy use / optimizes how space is used;

and minimizes the environmental impact of buildings;

8. [Maximum mark: 15]

22M.1.HL.TZ0.13

Cars have many automated features to improve the driving experience. One example is the use of headlights that automatically switch on and off.

(a) State **one** input device used by the automated headlights.

[1]

Markscheme

*Award [1 max]*

photoelectric sensor;

light sensor;

Examiners report

This question was reasonably answered except by weaker candidates

Some candidates failed to achieve full marks as result of incomplete and vague answers.

Some candidates demonstrated poor understanding of control systems.

Some cars have adaptive cruise control. This uses RADAR technology and a processor to ensure the car stays the same distance away from the car in front.

(b) Explain how the interaction between the inputs, processing and the outputs of the feedback loop ensure the car stays the same distance away from the car in front.

[6]

Markscheme

*Award [6 max]*

Sensor (continuously) collecting data (related to the distance between the two vehicles);

Data is converted to digital using ADC...;  
...and sent to the processor;  
Processor has access to pre-set data (minimum distance required between the two cars);  
Processor compares the input data against stored/pre-set data;  
If the vehicle is too close, the processor sends a signal to an actuator to apply the brakes/ an output signal (for example, warning light on the dashboard/ audible warning) to the driver to apply break;  
If the car has fallen back from the car in front, the processor sends a signal to an actuator to apply acceleration/ an output signal to the driver to apply acceleration;  
This process is constantly looped (feedback loop);

#### Examiners report

Many responses to part (b) were verbose while missing crucial information that would award the mark.

A system with many functions operating independently could be managed using either centralized or decentralized processing.

- (c) Discuss the use of centralized and decentralized processing in the context of control systems such as those involved in the operation of motor vehicles.

[5]

#### Markscheme

*Award [5 max]*

*Award [2 max] for centralised processing*

Centralized processing would allow the different features of the car to be controlled by a single processor;  
... this means that different features of the car are less likely to impact on the operation of other features, as the central processor can coordinate them;

*Award [2 max] for decentralised / distributed processing*

Decentralized processing allows each of the car's features to have its own processor;

... enabling each feature to function independently/more quickly;

*Award [1 max] for concluding statement*

Centralized processing is more complex for the manufacturer to program, but is possibly safer as the different components coordinate;

Centralised processing is more expensive to achieve as the whole system needs to be coordinated in real time;

Decentralised processing has a faster response (time due to the use of multiple smaller PLCs);

Decentralised processing is simpler/cheaper for the car manufacturer as they can use standard parts that are already programmed, but it is more difficult to get them to coordinate with each other;

Decentralised processing allows scalability, the subsystems can be easily customised;

A crash in a centralized system would result in failure of the whole system / a critical system failure so would have far greater consequences than in a decentralized system where the system may be able to partially function hence are riskier than decentralized ones;

As vehicles become fully autonomous (self-driving), ethical considerations must be taken into account when designing software.

- (d) Explain **one** ethical concern that must be considered in the development of autonomous vehicles.

[3]

Markscheme

*Award [3 max]*

Autonomous vehicles can be seen as more dangerous to pedestrians/cyclists;

...as they have to be programmed to react to avoid unpredictable movements from them;

The ethical consideration is how might the processor decide whether to avoid the cyclist to hit a pedestrian instead (trolley problem);

When autonomous vehicle control systems are perfected;

... they should be safer than human drivers;

... it may then be an ethical consideration as to whether a human who cannot drive should be the responsible person in the autonomous vehicle?;

An autonomous driving system could collect detailed information about the movements of the car;

... this information could reveal where individuals go, and raise privacy concerns;

... which could be exploited for commercial gain/ be considered as a form of surveillance (an ethical consideration);

There may be times when an accident is unavoidable (the trolley problem);

The autonomous vehicle has to make a 'decision' about who to hit;

In this case is the driver, the manufacturer, the software engineer or the third party accountable?;

*Note to examiners: accept other reasonable answers.*

9. [Maximum mark: 15]

21N.1.HL.TZ0.11

A rail transport company uses a global positioning system (GPS) to determine a train's position.

(a) Explain how GPS works.

[4]

Markscheme

*Award [4 max]*

The GPS receiver in a train takes the information from the satellite/ picks up the signals from (at least) 3 satellites;

The signals transmitted are: time of transmission, coordinates of the satellite;

The difference between the (atomic) time of transmission and (atomic) time of receiving the signal is used to calculate the distance;  
The method of trilateration is used to determine position from the distance to satellites/ is used to determine the train's exact position/ is used to calculate position of the train through equation resolution on a sphere;

#### Examiners report

Most candidates knew that a GPS works by communication with satellites. Some candidates failed to achieve full marks as result of incomplete and vague descriptions. A few candidates, for some unknown reasons, wrote that the satellite calculates the train's position.

The GPS data is used to provide real-time arrival data on video displays in train stations.

- (b.i) State **one** data item concerning the arrival of a train that may be provided on the video display.

[1]

#### Markscheme

*Award [1 max].*

Train number/ route / direction;  
Status (on time / delayed / cancelled);  
Time train / **next** train are due;

#### Examiners report

Generally, well answered, with many full or near full marks awarded.

- (b.ii) Outline how sensors can be used in combination with GPS to provide more accurate arrival data.

[2]

### Markscheme

*Award [2 max].*

Motion sensors/vibration sensors (can be placed on track rail);

*(Accept any sensors which detect trains that pass them!)*

(Sensing devices are placed) at a distance from each side of a train station/junction

OR

at some fixed points where obstacles (like tunnels) prevent GPS from working properly;

Data from each sensor can be logged continuously and used in calculations/determinations (of expected arrival times/ delays / rerouting during disruption, etc.);

### Examiners report

Generally, well answered, with many full or near full marks awarded.

The GPS data is made publicly available so that software developers can use it to build apps.

The mobile application *ATrainAway* uses the real-time train GPS data as well as the GPS data from the user's smartphone.

- (c) Outline **two** pieces of information that the *ATrainAway* application could provide to the user.

[4]

### Markscheme

*Award [4 max].*

Real-time information (train number/ route / direction and status (on time / delayed / cancelled);

relating to the arrival / departure of a particular train;

The time next train is due (the next 2/3 times a train will arrive at the station);

helps passenger to spend less time waiting on trains;

Should save user's (passenger) most frequented stops and routes;  
for quick access to get the next rail trip;

Alerts (set);  
to let you know when a train is nearing the station;

Trip planning (conducted);  
through Google trip planner within the app;

Information on the nearest station, distance to station/ next train arrival/  
next train departure;  
to help passenger to spend less time travelling/ save time;

*Marks 2 and 2.*

#### Examiners report

This question was well answered with the vast majority of candidates gaining some marks, some of whom achieved full or nearly full marks.

- (d) Discuss the advantages **and** disadvantages of using GPS in transportation systems.

[4]

#### Markscheme

*Award [4 max].*

Safer transit;

Trains are tracked and in case of emergency the location can be reached at the earliest and emergency services provided;

More favourable view of transit/helps passenger to get punctuality status of the train/;

GPS can provide worldwide three-dimensional positions (24 hours a day) (in any type of weather);

Sometimes it is too difficult to ensure reliable positioning;

Objects, such as buildings, overpasses, and other obstructions (that shield the antenna from a satellite) can potentially weaken a satellite's signal;

Privacy issues;

Knowing the absolute position of anything, anytime, anywhere;

*Marks 2 and 2.*

#### Examiners report

Candidates were usually able to identify the advantages and disadvantages of using GPS in transportation systems. Some marks were lost because of vague discussions.